Small Company IT Infrastructure: Overview

Practical assignment. This is a **proposed design**; nothing was deployed or tested. Assumptions: one office, 15 employees (about eight developers), Microsoft 365 and Azure tenants already in place, no on-premises servers. Full documentation, decisions and diagrams: **github.com/amirhosein2015/open-infrastructure-portfolio**

Part 1 — Infrastructure Overview

1. Employee computers and user accounts

Each employee has one **Microsoft Entra ID** account and one company-owned laptop. Access is given through **groups** (for example Staff, Developers), never person by person. Laptops follow a documented baseline: disk encryption with recovery keys stored safely, regular updates, antimalware, and no daily local administrator. Joiners and leavers follow a checklist; a leaver's sign-in is blocked first and sessions are revoked. Central device management (Intune) is an upgrade only if the licence includes it.

2. Office network and internet connectivity

One internet link feeds a **firewall**, a managed switch and Wi-Fi access points. Three VLANs: **Staff** (10), **Guest** (20, internet only) and **Management** (99, network devices). The firewall is the single policy point: unsolicited inbound traffic is denied and guests cannot reach staff or management networks. Staff Wi-Fi uses a shared passphrase; 802.1X is the upgrade path. There is no VPN: Microsoft 365 and Azure are reached over the internet and protected by identity (MFA), not by network location.

3. Microsoft 365 and Azure services

Microsoft 365 provides email, Teams, SharePoint and OneDrive. **Azure** hosts the applications in **one subscription** with one resource group per environment plus **rg-shared-ops** for monitoring (Log Analytics, alerts) and backup. Both services use the same Entra ID tenant. Azure permissions (Azure RBAC) are separate from Microsoft 365 roles and are assigned to groups at resource-group scope. The hosting and database products are deliberately not chosen: the scenario lacks the requirements to decide.

4. Development, staging and production environments

Each environment has its own resource group and its own **Key Vault** for secrets; applications read secrets through managed identities. Developers work freely in Development, have read-only access to Staging, and only logged read-only access to Production (no standing administration). Code goes through short-lived branches and review into a protected main branch. The pipeline **builds one artifact** and promotes it Development, then Staging, then Production only after approval by someone other than the requester. Deployments use a federated identity per environment, so no pipeline password is stored. Rollback means redeploying the previous artifact.

5. Security, access management and backup

Access: MFA through free security defaults, with Conditional Access when the licence allows. Administrators use separate admin accounts; two emergency accounts prevent lockout. Access is reviewed quarterly, and changes to administrator roles raise alerts. **Backup:** Azure Backup protects only supported Azure workloads and is **not** a Microsoft 365 backup. Microsoft 365 data relies on built-in recovery (recycle bins, version history, limited retention windows); a dedicated backup is deferred until the business sets recovery objectives. RPO and RTO are intentionally not invented here. A backup counts only after a successful **restore test**.

Part 2 — Infrastructure Diagram

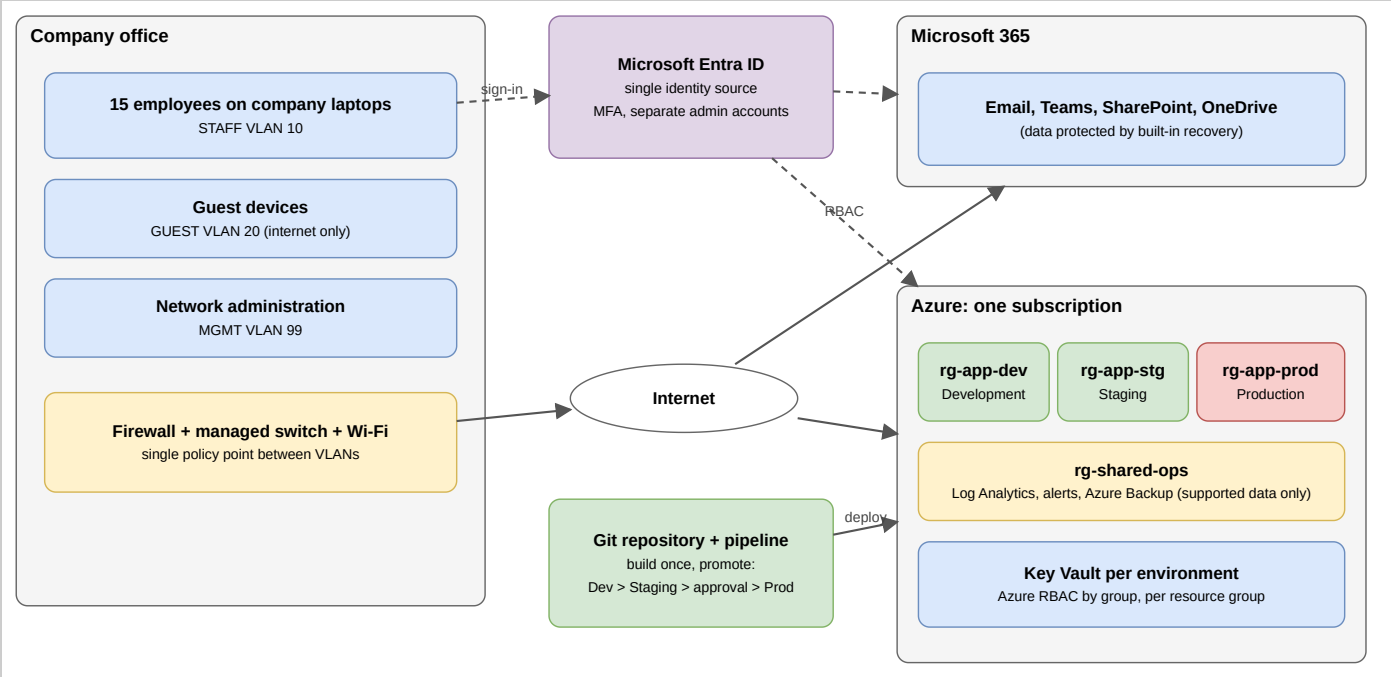

Diagram 1: main components and how the office, users, Microsoft 365, Azure and the three environments relate. Solid arrows are traffic, dashed arrows are identity.

Key decisions and trade-offs

Decision	Gain	Cost
Security defaults for MFA	Free, simple	Less control than Conditional Access
Three VLANs, one firewall policy point	Guests isolated; easy to explain	Less granular than many segments
Shared-passphrase Wi-Fi	Simple to run	Weaker than 802.1X; passphrase rotation on leavers
One subscription, resource group per environment	Low overhead	Logical, not complete, isolation
No VPN	Lower cost and complexity	No private network path to Azure
Build once, approve before Production	Tested artifact is deployed; human check	Slower releases

Limits: licence-dependent features (Conditional Access, Intune) are not assumed; recovery objectives, retention and thresholds are to be defined with the business.